

Strong Interlayer Coupling Accompanying Order Suppression in $\text{La}_{2/3}\text{Sr}_{1/3}\text{MnO}_3$ / $\text{La}_{1/3}\text{Sr}_{2/3}\text{FeO}_3$ Heterostructures

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Perovskite oxides are known for exhibiting a wide range of properties due to their coupled spin, lattice, charge, and orbital degrees of freedom, which could greatly enhance the control and tunability offered in oxide-based spintronics. Recent work has revealed novel low temperature behavior in the perovskite ferrite $\text{La}_{1/3}\text{Sr}_{2/3}\text{FeO}_3$ (LSFO), which is antiferromagnetic (AFM) and charge ordered (CO) below 190 K but undergoes a suppression of its known orders below 100 K. Working towards all-oxide spintronics, we investigate magnetic properties and interlayer coupling as a function of temperature and LSFO thickness in heterostructures, in which LSFO is layered with ferromagnetic (FM) $\text{La}_{2/3}\text{Sr}_{1/3}\text{MnO}_3$ (LSMO). We observe clear coupling between the FM LSMO and AFM LSFO in the form of an exchange bias, sharp increases in coercivity, and a jump in resistance at low temperature. Further, X-ray magnetic circular dichroism (XMCD) reveals a novel temperature dependence in LSMO/LSFO that is not present in individual films. Together, these measurements reveal that the overall heterostructure behavior is sensitive to both of LSFO's transition temperatures and that interlayer coupling induces novel behavior not present in single films. These results speak to the rich coupling present in perovskite oxide heterostructures and offer potential for greater tunability in spintronics applications.

Index Terms—antiferromagnetism, exchange bias, magnetic heterostructures, perovskites

I. INTRODUCTION

PEROVSKITE oxides are a rich class of materials known for their strong degree of coupling between lattice, charge, spin, and orbital degrees of freedom and the exotic phenomena that result from such strong coupling [1]–[4]. Due to their rich physics, perovskite oxides can potentially offer improved efficiency and tunability in spintronics applications by harnessing their strong correlations and coupled degrees of freedom.

One example of a strongly coupled perovskite is the mixed valence ferrite $\text{La}_{1/3}\text{Sr}_{2/3}\text{FeO}_3$ (LSFO). As shown in Fig. 1b, below $T_N \sim 190$ K, LSFO has coupled antiferromagnetism (AFM) and charge disproportionation along the $[111]_{pc}$ (pc = pseudocubic) direction with sixfold ($\uparrow\uparrow\uparrow\downarrow\downarrow\downarrow$) and threefold ($\text{Fe}^{3+}-\text{Fe}^{3+}-\text{Fe}^{5+}$) periodicity, respectively.

Using resonant X-ray scattering to study LSFO's ($\frac{1}{6} \frac{1}{6} \frac{1}{6}$) AFM peak and ($\frac{1}{3} \frac{1}{3} \frac{1}{3}$) charge order (CO) peak, we recently discovered that these coupled orders are suppressed at low temperature in LSFO thin films. As shown in Fig. 1c and d, respectively, the AFM and CO Bragg peaks appear at T_N , saturate in intensity, and subsequently decrease in intensity before disappearing completely below ~ 25 K. This disappearance was a new discovery not previously reported in LSFO thin films, and we proposed that in the low temperature regime, the low dimensionality of these LSFO thin films destabilizes the long-range magnetic and charge orders.

To further study and understand this novel low temperature behavior in LSFO, as well as how we can harness it for spintronics applications, we turn to LSFO-based heterostructures, in which LSFO is layered with the ferromagnetic (FM)

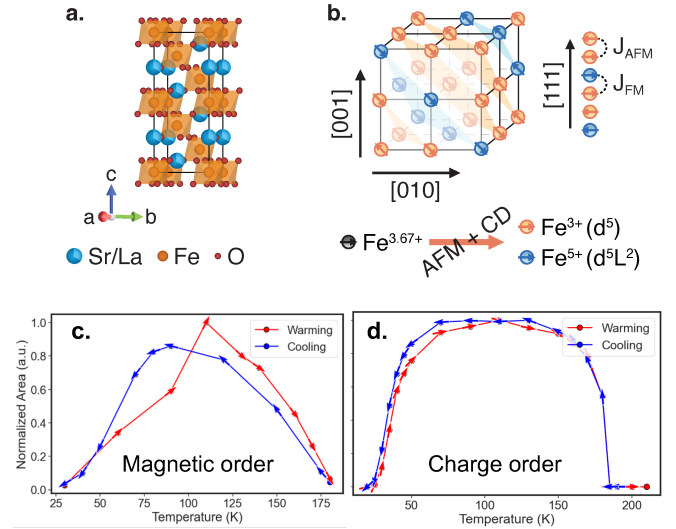


Fig. 1. a) Crystal structure and b) magnetic and charge order schematic for LSFO. c,d) Peak intensity versus temperature for LSFO's AFM and CO peaks, respectively, measured with resonant X-ray scattering.

mixed valence manganite $\text{La}_{2/3}\text{Sr}_{1/3}\text{MnO}_3$ (LSMO). LSMO has a Curie temperature above room temperature ($T_C \sim 360$ K), low damping [5], and half metallicity [6], making it a promising candidate for spintronics. These LSMO/LSFO heterostructures will provide insight into whether a neighboring FM layer stabilizes LSFO's order or, alternatively, whether the overall heterostructure behavior is affected by LSFO's order suppression.

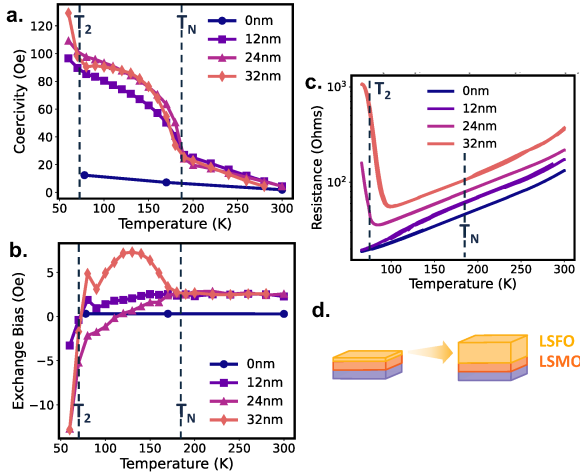


Fig. 2. Temperature dependence of a) coercivity, b) exchange bias, and c) resistance for LSMO(15 nm)/LSFO(x nm) heterostructures.

II. MAGNETOMETRY & TRANSPORT MEASUREMENTS

In this work, we study a series of LSMO/LSFO bilayers grown on LSAT(111) substrates with varying LSFO thicknesses. To determine the overall heterostructure's magnetic and transport behavior, we use vibrating sample magnetometry (VSM) and resistance measurements.

By taking moment vs field hysteresis loops throughout a temperature range spanning both of LSFO's transition temperatures, we extract the temperature dependence of the heterostructure's coercivity (Fig. 2a) and exchange bias (Fig. 2b). Compared to the bare LSMO film, the bilayers exhibit increased coercivity with clear kinks at LSFO's Néel temperature (T_N) and around the onset of LSFO's order suppression (T_2). Additionally, the heterostructures have an increasing exchange bias (EB) at low temperatures, with thicker LSFO layers corresponding to larger bias fields. The bilayer with the thickest LSFO layer further exhibits a positive exchange bias below T_N that switches sign below T_2 . These magnetometry measurements reveal strong magnetic coupling between the two materials, as indicated by the coercivity kinks and the nonzero exchange bias, and a possible thickness- and temperature-dependence to the interfacial coupling mechanism, as given by the change in EB sign for the thickest sample.

Heterostructure resistance was measured across a wide temperature range using a four-point Van der Pauw method. As shown in Fig. 2c, the heterostructures reflect LSMO's metallic behavior in the high and intermediate temperature regimes. However, around the suppression temperature T_2 , the bilayers' resistance sharply increases, deviating from the behavior of bare LSMO.

Together, these magnetometry and resistance measurements reveal that the LSMO/LSFO heterostructures are sensitive to and reflect both of LSFO's phase transitions in their magnetic and electronic behavior. This clear evidence of interlayer coupling motivates a further element-specific investigation to determine how each individual layer's properties differ in the heterostructure compared to in a bare film.

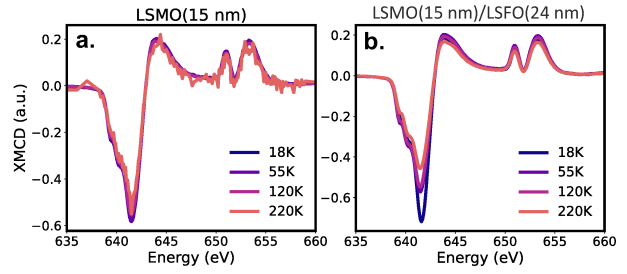


Fig. 3. Mn-edge XMCD measurements taken between 18 K and 220 K for a) a single LSMO film, and b) an LSMO/LSFO bilayer.

III. MN-EDGE X-RAY MAGNETIC CIRCULAR DICHROISM

X-ray magnetic circular dichroism (XMCD) measurements performed at the Mn $L_{2,3}$ edges allow us to isolate LSMO's magnetic behavior from the rest of the heterostructure and compare it to a bare LSMO film. As shown in Fig. 3, we measured Mn-edge XMCD in a bare LSMO film (Fig. 3a) and an LSMO/LSFO bilayer (Fig. 3b) at four temperatures spanning both of LSFO's transition temperatures. The magnitude of the bare LSMO XMCD is relatively constant across the entire temperature range from 18 K to 220 K. However, in the bilayer, the Mn edge XMCD exhibits a strong temperature dependence; the L_3 peak intensity clearly increases in magnitude across both T_N and T_2 . Therefore, the interfacial coupling between LSMO and LSFO in the bilayer induces a novel temperature dependence not seen in bare LSMO films.

IV. CONCLUSION

In conclusion, we observe clear interlayer coupling in LSMO/LSFO bilayers. A low temperature exchange bias and jump in resistance demonstrate that the heterostructures reflect LSFO's low temperature order suppression in their magnetic and electronic properties. Mn-edge XMCD reveals a novel temperature dependence induced in the LSMO layer of the heterostructures that is not present in single LSMO films, further showing the strong influence LSFO has on not only the overall heterostructure behavior but also on the LSMO layer specifically. This significant influence and coupling makes these heterostructures promising candidates for tunable oxide-based spintronics.

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